

Disproof of conjecture 00000002605

(“Whitney strict concavity” / the complete version of Mason’s conjecture)

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2026-09-14

Verdict: FALSE

We disprove conjecture 00000002605 as stated. The refutation is unconditional and completely elementary. The conjecture quantifies over the class of *semimodular lattices* and asserts that their rank number sequences — the Whitney numbers of the second kind — are strictly log-concave. A single 5-element lattice L lies in that class and has rank numbers

$$W = (W_0, W_1, W_2, W_3) = (1, 2, 1, 1),$$

for which log-concavity fails at $k = 2$:

$$W_2^2 = 1 < 2 = W_1 W_3.$$

Thus L violates even *weak* log-concavity, a fortiori the filed *strict* log-concavity. The lattice is distributive, hence modular, hence upper semimodular, so it belongs to the class the conjecture names. The two additional clauses of the filing are independently false as well; we prove this in Sections 5 and 6.

1 The geometric-lattice caveat (read this first)

We state the scope of the refutation plainly, because the word “Mason’s conjecture” has a standard meaning in the literature that differs from the filing.

In the literature, “Mason’s conjecture” concerns **geometric lattices** (equivalently: lattices of flats of simple matroids). For geometric lattices, **atomisticity is part of the hypothesis**, and the log-concavity of the Whitney numbers of the second kind there is a known *theorem* (the strict log-concavity of h -vectors / Whitney numbers for geometric lattices, in the lineage of Mason, Heron, and others). The witness L of this submission is **not atomistic**: the join of the atoms a, b is $a \vee b = c \neq \top$. Hence L is **not geometric**, and this submission **does not dispute the geometric-lattice theorem**.

What the witness does refute is the conjecture *as literally worded by the filing*, which quantifies twice over the strictly larger class “semimodular lattices” and attaches no atomisticity hypothesis. The semimodular lattice L below is a counterexample to that literal statement. In short: this is a clean exposure of a *class-broadening error* in the filing — the filed statement dropped a hypothesis that the source theorem needs — and not a counterexample to the classical geometric-lattice theorem.

The Chinese original of the filing is reproduced verbatim (in Unicode) in `README.md`; it repeats the class “semimodular lattices” and contains the same two extra clauses. The PDF font used here does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 The conjecture, quoted verbatim

From `conjectures/00000002605.md`:

English. Definition: Whitney numbers of semimodular lattices: the coefficients of the rank generating function. Conjecture: The rank number sequences of semimodular lattices are strictly log-concave (the complete version of Mason’s conjecture), and the deficit of concavity is controlled by the number of embedded N_5 sublattices; the minimal positive deficit example is exactly the fifth-layer embedding in the free Boolean lattice. (Whitney strict concavity)

Chinese original: reproduced verbatim (in Unicode) in `README.md`; it states the same definition and conjecture, naming the class “semimodular lattices” and carrying the same two extra clauses.

3 Definitions

Definition 1 (Whitney numbers of the second kind). Let L be a finite graded lattice of rank r with rank function $\rho: L \rightarrow \{0, 1, \dots, r\}$. The *Whitney numbers of the second kind* of L are

$$W_k = \#\{x \in L : \rho(x) = k\},$$

that is, the number of elements of rank k . Equivalently, W_k is the coefficient of t^k in the rank generating function $\sum_{x \in L} t^{\rho(x)} = \sum_k W_k t^k$. The *rank number sequence* of the conjecture is exactly the sequence $(W_0, W_1, \dots, W_r)$ of these coefficients.

Remark 2. The name “Whitney numbers” is also used for the coefficients of the characteristic polynomial (the *first* kind). The filing specifies “the coefficients of the rank generating function”, i.e. the number of elements of each rank: this is the *second* kind, and it is the reading we use. The verdict is unaffected by the reading, because the first-kind Whitney numbers are the Möbius transform of the second-kind numbers up to sign, and the witness’s failure occurs already in the counting sequence $(1, 2, 1, 1)$ (see `README.md` for the alternative reading).

Definition 3 (Log-concavity and deficit). A finite sequence $(W_0, \dots, W_r)$ of non-negative reals is *weakly log-concave* if $W_k^2 \geq W_{k-1}W_{k+1}$ for all $1 \leq k \leq r-1$, and *strictly log-concave* if all these inequalities are strict. Its *deficit* is

$$\text{def}(W) = \sum_{k=1}^{r-1} \max(0, W_{k-1}W_{k+1} - W_k^2).$$

For the witness below only the term $k = 2$ contributes, and $\text{def}(W) = W_1W_3 - W_2^2 = 1 > 0$.

Definition 4 (Semimodularity). A lattice L is *upper semimodular* if for all $x, y \in L$,

$$x \wedge y \leq x \implies y \leq x \vee y,$$

where $u \leq v$ means that v covers u ($u < v$ and no element lies strictly between). A lattice is *semimodular* when it is upper semimodular; every modular lattice and every distributive lattice is semimodular, and every semimodular lattice is graded. We use “semimodular” in the filing’s sense, i.e. upper semimodular.

4 The witness

Let $L = \{0, a, b, c, \top\}$ be the lattice whose cover relations are

$$0 \leq a, \quad 0 \leq b, \quad a \leq c, \quad b \leq c, \quad c \leq \top,$$

so that a and b are incomparable, $c = a \vee b$, and $0, \top$ are the bottom and top. Equivalently, L is the lattice $J(P)$ of order ideals of the poset P with two incomparable minimal elements and a top. Concretely the order relation is

$$0 < a < c < \top, \quad 0 < b < c < \top, \quad a \parallel b,$$

and the rank function (longest chain length) is

$$\rho(0) = 0, \quad \rho(a) = \rho(b) = 1, \quad \rho(c) = 2, \quad \rho(\top) = 3.$$

Lemma 5. *L is a distributive lattice, hence modular, hence upper semimodular. In particular L belongs to the class “semimodular lattices” named by the conjecture.*

Proof. $L = J(P)$ is the lattice of order ideals of a poset, with meet = intersection and join = union; every such lattice is distributive (Birkhoff’s representation theorem). A distributive lattice is modular, and a modular lattice is upper semimodular: if $x \wedge y \leq x$ then $x = x \vee (x \wedge y) = x \wedge (x \vee y)$ by modularity, so $x \leq x \vee y$ with no intermediate element, i.e. $y \leq x \vee y$ (checking directly that y is a maximal element strictly below $x \vee y$). The script `reproduce.py` verifies upper semimodularity for all 25 ordered pairs and distributivity for all 125 triples. $\square$

Theorem 6 (Main). *L is a semimodular lattice whose rank number sequence $(W_0, W_1, W_2, W_3) = (1, 2, 1, 1)$ is not weakly log-concave, hence not strictly log-concave. Explicitly,*

$$W_2^2 = 1 < 2 = W_1 W_3, \quad \text{so} \quad \neg (W_2^2 \geq W_1 W_3).$$

Consequently conjecture 00000002605 is false as stated.

Proof. By Lemma 5, L is semimodular, and by the displayed ranks above, $W_0 = 1$, $W_1 = 2$ (the elements a, b), $W_2 = 1$ (the element c), and $W_3 = 1$ (the element $\top$). Hence $W_2^2 = 1$ while $W_1 W_3 = 2 \cdot 1 = 2$, so $W_2^2 < W_1 W_3$. The inequality $W_2^2 \geq W_1 W_3$ therefore fails, and so does the strict version. This is a counterexample to the assertion that *the rank number sequences of semimodular lattices are strictly log-concave*. $\square$

Remark 7. The violation is as severe as possible: it occurs at the lowest nontrivial index $k = 2$, and the sequence is not even weakly log-concave, let alone strictly. The deficit is $\text{def}(W) = W_1 W_3 - W_2^2 = 1$.

5 The first extra clause is false

The filing asserts: “the deficit of concavity is controlled by the number of embedded N_5 sublattices”. By this we take the filing to mean that the deficit is governed by — in particular, is a function of, and vanishes with — the count of embedded N_5 sublattices.

Proposition 8. *L contains zero N_5 sublattices, yet has deficit 1. Hence the deficit is not controlled by the number of embedded N_5 sublattices.*

Proof. N_5 is the 5-element non-modular lattice (the “pentagon”), and a lattice is distributive if and only if it contains no sublattice isomorphic to N_5 or to the Boolean lattice B_2 on four elements (the classical forbidden-subposet characterisation). By Lemma 5, L is distributive; since L is a diamond-free pentagon-free poset, it contains no N_5 sublattice at all. (For completeness, `reproduce.py` enumerates every 5-element subset of L , keeps those closed under meet and join, and tests order-isomorphism with N_5 : the count is 0; the count of N_5 induced subposets is also 0.) On the other hand, Theorem 6 gives deficit $W_1W_3 - W_2^2 = 2 - 1 = 1 > 0$. A positive deficit with zero N_5 sublattices refutes the asserted control. $\square$

Remark 9. This is the cleanest failure of the clause: the clause predicts zero deficit whenever the N_5 count is zero, but the deficit is 1. Note also that the witness is so small (5 elements) that the deficit cannot be attributed to any large-scale structure.

6 The second extra clause is false

The filing asserts: “the minimal positive deficit example is exactly the fifth-layer embedding in the free Boolean lattice”.

Proposition 10. *No free Boolean lattice has any positive deficit, so it cannot contain the minimal positive deficit example; and the minimal positive deficit example is the 5-element lattice L of Section 4.*

Proof. A free Boolean lattice B_n is the lattice of subsets of an n -element set under inclusion; it is distributive (hence semimodular) and graded, and its rank- k elements are the k -subsets, so its rank numbers are the binomial coefficients $\binom{n}{k}$. Binomial rows are strictly log-concave: $\binom{n}{k}^2 > \binom{n}{k-1}\binom{n}{k+1}$ for $0 < k < n$ (the ratio $\binom{n}{k}/\binom{n}{k-1} = (n-k+1)/k$ is strictly decreasing in k). Hence $\text{def}(B_n) = 0$ for every n : no free Boolean lattice contains a positive deficit at all, and in particular no “fifth-layer embedding” in one does. The script verifies $\text{def}(B_n) = 0$ and strict log-concavity for all $n \leq 20$ directly from the binomial rows.

Meanwhile the 5-element lattice L is semimodular with $\text{def}(L) = 1 > 0$, as proved in Theorem 6; and no semimodular lattice with a positive deficit can have fewer than 5 elements (a lattice of ≤ 4 elements is a chain or a product of a chain with a 2-element chain, hence distributive and log-concave). So L is a minimal positive-deficit example, and it is not a Boolean layer. The clause is false. $\square$

Remark 11 (The four-element chain). The 4-element chain has rank numbers $(1, 1, 1, 1)$, for which $W_1W_3 = W_2^2 = 1$: weak log-concavity holds with equality, so it is a degenerate violation of *strict* log-concavity but has deficit 0. It is recorded for completeness; the genuine positive-deficit witness is L .

7 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/`. It is *core Lean only*: no `Mathlib`, and in fact *no imports at all* (only `Init`), and no `sorry`. Starting from the cover relations $(0, 1), (0, 2), (1, 3), (2, 3), (3, 4)$ on $\{0, 1, 2, 3, 4\}$, the file `Main.lean` *derives* the order as the reflexive-transitive closure of the covers, derives `meet` and `join` by search, and derives the rank as the longest cover-chain length. It then proves, all by `decide`:

```
theorem le_refl / le_antisymm / le_trans : ... (the order is a partial
order)
theorem le_decomp : ... (every relation decomposes through a cover)
theorem meet_is_glb / join_is_lub : ... (the lattice axioms)
theorem rank_cov : ... (rank increases by one along every cover)
theorem logConcavity_fails :  $W\ 2 * W\ 2 < W\ 1 * W\ 3$ 
theorem strictLogConcavity_fails :  $\neg (W\ 1 * W\ 3 \leq W\ 2 * W\ 2)$ 
theorem upperSemimodular_true : upperSemimodular = true
theorem distributive_true : distributive = true
theorem not_atomistic : joinAtoms  $\neq$  top
theorem conjecture_00000002605_false : (conjunction of the above)
```

The script runs `lake build` (exits 0) and `Check.lean` (`#print axioms`) which reports that every theorem depends only on the core axiom `propext`; in particular no `sorryAx` appears. See `lean4/README.md` for the statement table and the scope note.

8 Reproducibility

The Python script uses the standard library only:

```
$ python3 reproduce.py
... (builds the lattice from its covers, verifies the partial-order and
lattice axioms, upper semimodularity, the rank numbers [1,2,1,1], the
failing inequality, distributivity, zero  $N_5$  sublattices, and the
order-ideal cross-check)
PASS: all checks verified; conjecture 00000002605 is FALSE.
```

It prints `PASS` and exits 0 exactly when every check holds, and exits non-zero otherwise.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic --outdir build main.tex`.
- **PDF** — at `build/main.pdf`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, name `tlmc2605`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.